Question 2.

A1 Workable arrangement and correct kit **1**
Do not allow this mark if the method of investigation is invalid.
E.g. Do not allow d.c. with magnet or d.c. with vibration generator.
Do not allow stopwatch methods
Do not allow if no method given for changing frequency
Do not allow if no method of finding tension (e.g. turning pegs on a guitar)
Allow microphone + CRO for frequency measurement
Allow tuning fork placed on sonometer + paper rider on wire

A2 Procedure (determine T and f, change T and measure new f) **1**
This mark may be scored even if the arrangement is unworkable
Could be given for table headings. If l varies then this mark cannot be scored.

B1 Method of measuring tension (e.g. weight of mass on wire/Newton meter) **1**
Do not accept methods involving measurement of extension

B2 Method of measuring frequency (e.g. read scale from SG or strobe) **1**
Allow microphone connected to CRO to be placed near to the wire and 'find the frequency from the CRO'. Allow frequency from the fork to be given.
Allow use of 'high speed photography' if sensible
Do not accept 'frequency sensor'. Do not accept 'camera' or 'video'

C1 Method of achieving resonance of wire **1**
(e.g. adjust output freq. of SG until large amplitude oscillations are seen)
Allow 'pluck the wire'

C2 Any one safety precaution **1**
(e.g. wear safety goggles (in case the wire snaps); safety screens; keep feet away from the load in case the wire snaps; use a bucket of sand below load; do not look directly into strobe). Do not allow 'gloves'.

C3 Use constant length of wire or same mass per unit length of wire **1**

D1/2/3 Any further good design features. Some of these might be: **3**
Place white card behind vibrating wire so it can be seen easily.
Check frequency scale of SG/calibration ideas
Good description for finding resonance position
Use top pan balance/electronic scales to find mass
Do preliminary experimentation to determine suitable loads
Detail relating to how the frequency would be found from the CRO trace
Check uniformity of wire (1 mark) Use screw gauge to check the uniformity (1 mark)
Sensible ideas relating to the elastic limit
Perform the experiment in a quiet place if using microphone and CRO
Check the wire to ensure that it is free from kinks
Accept 'smooth pulley' or 'smooth rod' ideas
Do not accept 'repeat readings' ideas

10 marks in total.

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